[1]

```python
import numpy as np
subjects = ["Math", "Science", "English", "History"]
student_scores = []
for i in range(4):
    marks = list(map(int, input(f"Enter marks for Student {i+1} (Math Science English History): ").split()))
    student_scores.append(marks)
student_scores = np.array(student_scores)
subject_averages = np.mean(student_scores, axis=0)
highest_index = np.argmax(subject_averages)
print("\nAverage score for each subject:")
for i in range(4):
    print(subjects[i], ":", subject_averages[i])
print("\nSubject with highest average:", subjects[highest_index])
print("Highest average score:", subject_averages[highest_index])
```

```
Enter marks for Student 1 (Math Science English History): 90 80 78 88
Enter marks for Student 2 (Math Science English History): 90 90 90 90
Enter marks for Student 3 (Math Science English History): 90 98 98 78
Enter marks for Student 4 (Math Science English History): 98 98 89 99

Average score for each subject:
Math : 92.0
Science : 91.5
English : 88.75
History : 88.75

Subject with highest average: Math
Highest average score: 92.0
```

```python
import numpy as np
sales = []
print("Enter the prices of 3 products:")
for i in range(3):
    row = list(map(float, input("Product " + str(i + 1) + ": ").split()))
    sales.append(row)
sales = np.array(sales)
average_price = np.mean(sales)
print("\nSales data:")
print(sales)
print("\nAverage price of all products:", average_price)
```

```
Enter the prices of 3 products:
Product 1: 100 800 780
Product 2: 890 900 888
Product 3: 876 560 456

Sales data:
[[100. 800. 780.]
 [890. 900. 888.]
 [876. 560. 456.]]

Average price of all products: 694.4444444444445
```

```python
import numpy as np
house_data = []
n = int(input("Enter number of houses: "))
print("Enter data in this order: Bedrooms Square_Footage Sale_Price")
for i in range(n):
    data = list(map(float, input("House " + str(i + 1) + ": ").split()))
    house_data.append(data)
house_data = np.array(house_data)
houses = house_data[house_data[:, 0] > 4]
if len(houses) > 0:
    average_price = np.mean(houses[:, 2])
    print("\nAverage sale price of houses with more than 4 bedrooms:",
          average_price)
else:
    print("\nNo houses with more than 4 bedrooms.")
```

```
Enter number of houses: 2
Enter data in this order: Bedrooms Square_Footage Sale_Price
House 1: 5 1500 250000
House 2: 4 5699 450000

Average sale price of houses with more than 4 bedrooms: 250000.0
```

```python
import numpy as np
sales_data = []
for i in range(4):
    sales = float(input("Enter sales for Quarter " + str(i + 1) + ": "))
    sales_data.append(sales)
sales_data = np.array(sales_data)
total_sales = np.sum(sales_data)
percentage_increase = ((sales_data[3] - sales_data[0]) / sales_data[0]) * 100
print("Sales data:", sales_data)
print("Total sales for the year:", total_sales)
print("Percentage increase from Q1 to Q4:", percentage_increase, "%")
```

ndarray: sales_data

ndarray with shape (4,)

```
Enter sales for Quarter 1: 340000
Enter sales for Quarter 2: 560000
Enter sales for Quarter 3: 560000
Enter sales for Quarter 4: 430000
Sales data: [340000. 560000. 560000. 430000.]
Total sales for the year: 1890000.0
Percentage increase from Q1 to Q4: 26.47058823529412 %
```

```python
import numpy as np
n = int(input("Enter number of car models: "))
fuel_efficiency = []
for i in range(n):
    efficiency = float(input("Enter fuel efficiency for Car Model " + str(i + 1) + ": "))
    fuel_efficiency.append(efficiency)
fuel_efficiency = np.array(fuel_efficiency)
average = np.mean(fuel_efficiency)
print("Average fuel efficiency:", average)
car1 = float(input("Enter fuel efficiency of first car: "))
car2 = float(input("Enter fuel efficiency of second car: "))
percentage_improvement = ((car2 - car1) / car1) * 100
print("Percentage improvement:", percentage_improvement, "%")
```

```
Enter number of car models: 4
Enter fuel efficiency for Car Model 1: 45
Enter fuel efficiency for Car Model 2: 30
Enter fuel efficiency for Car Model 3: 20
Enter fuel efficiency for Car Model 4: 25
Average fuel efficiency: 30.0
Enter fuel efficiency of first car: 22
Enter fuel efficiency of second car: 28
Percentage improvement: 27.27272727272727 %
```

```python
prices = []
quantities = []
n = int(input("Enter number of items: "))
for i in range(n):
    price = float(input("Enter price of item " + str(i + 1) + ": "))
    quantity = int(input("Enter quantity of item " + str(i + 1) + ": "))
    prices.append(price)
    quantities.append(quantity)
discount_rate = float(input("Enter discount rate (%): "))
tax_rate = float(input("Enter tax rate (%): "))
subtotal = 0
for i in range(n):
    subtotal = subtotal + (prices[i] * quantities[i])
discount = subtotal * discount_rate / 100
price_after_discount = subtotal - discount
tax = price_after_discount * tax_rate / 100
total_cost = price_after_discount + tax
print("Subtotal:", subtotal)
print("Discount:", discount)
print("Price after discount:", price_after_discount)
print("Tax:", tax)
print("Total cost:", total_cost)
```

```
Enter number of items: 4
Enter price of item 1: 28
Enter quantity of item 1: 78
Enter price of item 2: 56
Enter quantity of item 2: 90
Enter price of item 3: 78
Enter quantity of item 3: 56
Enter price of item 4: 35
Enter quantity of item 4: 67
Enter discount rate (%): 50
Enter tax rate (%): 4
Subtotal: 13937.0
Discount: 6968.5
Price after discount: 6968.5
Tax: 278.74
Total cost: 7247.24
```